

ASHUTOSH SHARMA

AI Engineer | Agentic AI & RAG Systems

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SUMMARY

AI Engineer with 4+ years building agentic RAG systems that run in production for enterprise clients. Ownership runs end-to-end — ingestion, hybrid retrieval over pgvector, agent orchestration, and the Kubernetes and Celery layer underneath — working alongside domain experts and cross-functional delivery teams. Recent platform work doubled grounded-answer accuracy and cut p95 latency 35%. Also maintains pi-agent, an MIT-licensed CLI agent published on PyPI.

TECHNICAL SKILLS

GenAI & Agents: LLMs, Agentic AI, Multi-Agent Systems, ReAct Agents, Tool Use / Function Calling, LangGraph, LangChain, LlamaIndex, CrewAI, Google ADK, Microsoft Agent Framework / AutoGen, MCP (Model Context Protocol), A2A Protocol, Context Engineering, Structured Output, Prompt Engineering

Vector Search & RAG: RAG (Agentic, Hybrid, CRAG), Vector Databases, pgvector, FAISS, Weaviate, Embeddings, Hybrid Dense + Sparse Retrieval (BM25), Cohere Reranking, Query Decomposition, Grounded Citation, Chunking Strategies, Sentence Transformers, CLIP (Multimodal)

LLM Safety, Privacy & Evaluation: Guardrails, PII Detection / Redaction & Masking, Data Anonymization, GDPR Compliance, Prompt-Injection Defense, Multi-Tenant Isolation, Sandboxed Agent Execution (gVisor), RAGAS, LLM-as-Judge, LLM Observability, Phoenix Tracing, Prometheus

Backend, Cloud & DevOps: Python (async), FastAPI, PostgreSQL, SQL, Pydantic, SQLAlchemy, Celery, RabbitMQ, WebSocket & SSE Streaming, REST APIs, Microservices, Docker, Kubernetes, Helm, CI/CD, Cloud Deployment, AWS S3 / Object Storage, Git

ML & Data Science: PyTorch, Hugging Face Transformers, scikit-learn, LightGBM, XGBoost, LSTM, Random Forest, LDA Topic Modeling, Feature Engineering, ETL Pipelines, Pandas, NumPy

PROFESSIONAL EXPERIENCE

AI Engineer — Generative AI (Data Scientist) | T-Systems ICT India Pvt. Ltd.

December 2022 – Present

AI Operator — Agentic RAG Platform

May 2025 – Present

- Own a multi-node agentic RAG platform on LlamaIndex, FastAPI and pgvector serving enterprise document Q&A. Cut p95 response time by 35% under production load.
- Rebuilt citation handling around one source-numbering authority, so answers return only the documents they actually cite. Replaced a separate citation LLM pass that kept disagreeing with the source filter, cutting an LLM call per query.
- Introduced BM25 keyword search alongside pgvector with Cohere reranking on top, so queries hinging on an exact string (error codes, product IDs) stopped falling through semantic-only retrieval.
- Raised answer faithfulness from 40% to over 80%, measured on a RAGAS harness running 1,000+ evaluations a week with Phoenix tracing behind it. Presented the results to the VP of Engineering.
- Extended ingestion from PDF-only to six formats, partnering with cross-functional teams to onboard 15+ enterprise data sources.
- Deployed on Kubernetes with Helm at 99.5% uptime; Celery workers are memory-bounded because LlamaIndex objects leak across tasks.

Insurance Healthcare Agent — Agentic RAG System

December 2024 – April 2025

- Designed an agentic system for insurance support automation, answering from policy documents and PostgreSQL in one retrieval pass, with function calling to pick the source. Query-resolution efficiency up 40%.
- Optimized the agent workflows and their tools with domain experts, then ran them in production at 10,000+ queries a month, 98% accuracy, 99.5% uptime, with fallback handling when a tool call fails. Agent workload dropped 50%, wait times 70%.

CrewAI Multi-Agent Content System · Topic Modeling & Document Clustering

February 2024 – November 2024

- Orchestrated a writer/researcher/editor CrewAI pipeline across five use cases and cut blog production time by 80%.
- Clustered 1,000+ enterprise documents with LDA, then trained a LightGBM classifier to 90% accuracy on the extracted topics. Removed 70% of manual categorization.

Enterprise Data Analytics & ML Modeling

December 2022 – January 2024

- Modeled customer segmentation and classification with XGBoost and Random Forest at 87% accuracy. Built ETL pipelines over 50K+ records with feature engineering, cutting data-prep time 60%, and shipped the reporting dashboards for client stakeholders on top.

Data Science Intern | Analytics Labs

November 2021 – November 2022

- Developed a Random Forest model for customer attrition; targeted-segment attrition fell 12%. Established the KPI set for real-time CPE tracking across 500+ devices.

OPEN SOURCE

pi-agent — open-source coding agent

pypi.org/project/pi-coding-agent

- Published on PyPI (v0.6.0, MIT). Multi-provider — Claude, GPT, Gemini, Groq, local Ollama — with an MCP client, 21 packaged skills, sub-agents, persistent memory, and sandboxed tool execution behind guardrails. 174 tests in CI.

EDUCATION

Master of Science, Data Science — Manipal Academy of Higher Education, Manipal

July 2022 – July 2024

Bachelor of Science, Computer Science & Statistics — Bangalore North University

July 2018 – November 2021

Certifications: Generative AI with Large Language Models (DeepLearning.AI) · LangChain for LLM Application Development (DeepLearning.AI) · Machine Learning Specialization (Coursera) **Languages:** English (professional), Hindi (native)